

Customer Churn Analysis using Python & SQL Server

Executive Summary

This project analyzes customer churn using SQL Server and Python to identify retention opportunities through data cleaning, feature engineering, KPI analysis and visualizations.

Business Problem

Subscription businesses lose recurring revenue when customers churn. The goal is to identify patterns associated with churn and provide actionable recommendations.

Objectives

The main objectives of this project are to:

- Calculate the overall customer churn and retention rates.
- Analyse churn across different subscription plans, contract types, and geographical locations.
- Evaluate revenue generated by different customer segments.
- Measure the relationship between customer support escalations and churn.
- Create business-ready KPIs and visualizations for decision-making.
- Identify high-risk customer segments and recommend effective retention strategies.

Dataset

Three source tables: Customer, Subscription and Support. Joined using Customer ID after importing from SQL Server.

SQL Server Integration

Instead of directly analyzing Excel files, the raw data was first imported into SQL Server as separate relational tables. Python was then connected to SQL Server using the **PyODBC** library to retrieve data through SQL queries. After loading the data into Pandas Data Frames, the analysis was carried out within Jupyter Notebook.

Data Cleaning

Several preprocessing steps were performed to improve data quality before analysis:

- Renamed columns for better readability.
- Removed unnecessary columns such as interests, pin code, comment, and col_1.
- Converted date columns into appropriate datetime formats.
- Standardized categorical values (for example, converting inconsistent gender values).
- Filled missing country values using state-to-country mapping.
- Removed fully empty records.
- Identified and removed duplicate support records before merging tables.
- Validated the cleaned dataset before proceeding with analysis.

These steps ensured that the final dataset was accurate, consistent, and suitable for business analysis.

Feature Engineering

To enhance the analytical value of the dataset, several new features were created:

- Created a **churn_flag** column using the cancellation date, where cancelled customers were assigned a value of 1 and active customers a value of 0.
- Calculated customer tenure using subscription start and cancellation dates.
- Generated complaint count for each customer.
- Created customer churn risk categories (Low, Medium, High) using churn score.
- Converted Boolean escalation values into numerical format for correlation analysis.
- Prepared an encoded dataset for heatmap and correlation analysis.

These engineered features enabled more meaningful KPI calculations and advanced customer segmentation.

KPIs

The following business KPIs were calculated during the analysis:

- Overall Customer Churn Rate
- Customer Retention Rate

- Churn Rate by Plan Type
- Churn Rate by State
- Revenue by State
- Revenue by Subscription Type
- Customer Count by State
- Average Revenue Per User (ARPU)
- Average Customer Tenure
- Revenue at Risk
- Escalation Rate
- Average Complaint Rate
- Correlation between Escalations and Churn

These KPIs provide a comprehensive overview of customer behaviour, financial performance, and operational efficiency.

EDA

Exploratory Data Analysis was performed to identify trends and customer behaviour patterns across different business dimensions. GroupBy operations, aggregations, pivot tables, and statistical summaries were used to compare customer segments.

Several visualizations were created, including:

- Monthly Churn Trend
- Churn by Plan Type
- Churn by State
- Correlation Heatmap
- Pivot Table Analysis
- Revenue Distribution
- Customer Segmentation

These visualizations helped identify the factors most strongly associated with customer churn and provided a clear understanding of customer behaviour across different subscription plans and geographical regions.

Key Insights

- Churn Rate: 28.6% | Retention Rate: 71.4%
- Most of the churn is from basic subscription plan – nothing to worry in terms of major revenue impact
- Most of the churn happened in the month of Sep 2024 and, most affected state is Karnataka
- Average Tenure (Days): 1,451 | ARPU is Rs 18.8
- Total Revenue = 395
- Revenue loss due to churn = 74 | CLTV Lost = 2,047
- % Revenue loss = 18%
- monthly vs annual churn = 55.6% vs 8.3%

Action Items

- Check what happened in Karnataka – was there any increase in price, user complaints, tech issues, etc
- Did we increase subscription price for Basic plan or recently made any changes – specially in september month?
- Check what competitor is doing – because one user has moved to competitor
- Focus on customers with 'High' & 'Med' churn risk, check their LTV (to make priority list), complaint request, reach out to them through email, SMS, calls and fix their issue.

Business Recommendations

Based on the analysis, the following recommendations can help improve customer retention and reduce revenue loss:

- Launch proactive retention campaigns targeting customers identified as High and Medium churn risk.
- Encourage customers to migrate from monthly to annual subscription plans through discounts or loyalty programs.

- Improve customer support processes by reducing escalation rates and resolving customer issues more efficiently.
- Continuously monitor churn KPIs using interactive dashboards to detect changes in customer behaviour.
- Focus retention initiatives on high-value customer segments and regions that contribute the most revenue.
- Regularly review pricing strategies, subscription plans, and customer feedback to improve overall customer satisfaction and reduce churn.

Conclusion

This project demonstrates a complete end-to-end customer churn analysis workflow using SQL Server and Python. Starting from relational database integration, the project covers data cleaning, feature engineering, KPI generation, exploratory data analysis, visualization, and business recommendation development. Beyond technical implementation, the project emphasizes translating analytical findings into business decisions that can improve customer retention, optimize subscription strategies, and support long-term revenue growth.